# Gaussian and Nonlinear

$X(t)$ 为一个 GP, $\underbrace{Y(t)}_{\text{Non-Gaussian}} = g(X(t))$, 其中 $g$: Nonlinear, No memory.

$E(g(t))$ $\quad R_Y(t,s) = E(Y(t)Y(s))$

## ① Square 非线性:

$X(t)$ 为 GP, $E(X(t)) = 0$, w.s.s. $\qquad Y(t) = X^2(t) \geq 0$ 为一个 Non-GP

$E(Y(t)) = E(X^2(t)) = R_X(0)$

$R_Y(t,s) = E(Y(t), Y(s)) = E(X^2(t)\, X^2(s))$

$\longrightarrow$ 假设 $(X_1, X_2, X_3, X_4) \sim N$, $E(X_k) = 0$,

$$E(X_1X_2X_3X_4) = E(X_1X_2)E(X_3X_4) + E(X_1X_3)E(X_2X_4) + E(X_1X_4)E(X_2X_3)$$

Pf: 利用 Characteristic Function 证明.

$(X_1, \cdots, X_n)^T = X$, $\Phi_X(w) = E(\exp(jw^TX))$

$$= E[\exp(j(w_1X_1 + \cdots + w_nX_n))]$$

$\Longrightarrow E(X_1^{\alpha_1} \cdots X_n^{\alpha_n})$

$$= \frac{1}{j^{\alpha_1+\cdots+\alpha_n}} \frac{\partial^{\alpha_1+\cdots+\alpha_n}}{\partial w_1^{\alpha_1} \cdots \partial w_n^{\alpha_n}} \Phi_X(w) \Bigg|_{w_1=\cdots=w_n=0} \qquad \#$$

$$E(X_1X_2X_3X_4) = \frac{1}{j^4} \frac{\partial^4}{\partial w_1 \cdots \partial w_4} \Phi_X(w) \Bigg|_{w_1=\cdots=w_4=0}$$

$\square$

回到 $E(X^2(t)\,X^2(s)) = E(X^2(t))\,E(X^2(s)) + 2E(X(t)X(s))\,E(X(t)X(s))$

$$= R_X^2(0) + 2R_X^2(t-s)$$

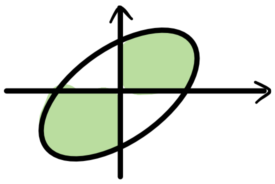
## ② Hard Limiter (Polar)

$X(t)$ 为 GP, 且 $E(X(t)) = 0$ $\qquad Y(t) = \operatorname{sgn}(X(t)) = \begin{cases} 1, & X(t) > 0 \\ -1, & X(t) < 0 \end{cases}$

$E(Y(t)) = 1 \cdot P(Y(t) = 1) + (-1) \cdot P(Y(t) = -1)$

$\qquad = 1 \cdot P(X(t) > 0) + (-1) \cdot P(X(t) < 0) = 0$

$R_Y(t,s) = E(Y(t)Y(s)) = 1 \cdot P(X(t)X(s) > 0) + (-1) \cdot P(X(t)X(s) < 0)$



$$f_{X(t),X(s)}(x_1,x_2) = \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} \exp\left(-\frac{1}{2(1-\rho^2)}\left(\left(\frac{x_1}{\sigma_1}\right)^2 + \left(\frac{x_2}{\sigma_2}\right)^2 - 2\rho\frac{x_1}{\sigma_1}\frac{x_2}{\sigma_2}\right)\right)$$

$$I = \left(\int_0^\infty\int_0^\infty + \int_{-\infty}^0\int_{-\infty}^0\right) \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} \exp\left(-\frac{1}{2(1-\rho^2)}\left(\left(\frac{x_1}{\sigma_1}\right)^2 + \left(\frac{x_2}{\sigma_2}\right)^2 - 2\rho\frac{x_1}{\sigma_1}\frac{x_2}{\sigma_2}\right)\right) dx_1dx_2$$

$$= \cancel{2} \int_0^\infty\int_0^\infty \frac{1}{\cancel{2}\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} \exp\left(-\frac{1}{2(1-\rho^2)}\left(\left(\frac{x_1}{\sigma_1}\right)^2 + \left(\frac{x_2}{\sigma_2}\right)^2 - 2\rho\frac{x_1}{\sigma_1}\frac{x_2}{\sigma_2}\right)\right) dx_1dx_2$$

取 $y_1 = \dfrac{x_1}{\sqrt{1-\rho^2}\,\sigma_1}$, $y_2 = \dfrac{x_2}{\sqrt{1-\rho^2}\,\sigma_2}$ $\Rightarrow dx_1dx_2 = (1-\rho^2)\sigma_1\sigma_2\,dy_1dy_2$

故 $I = \displaystyle\int_0^\infty\int_0^\infty \frac{\sqrt{1-\rho^2}}{\pi} \exp\left(-\frac{1}{2}(y_1^2 + y_2^2 - 2\rho y_1y_2)\right) dy_1dy_2$

取 $y_1 = u+v, \quad y_2 = u-v$

$$I = \iint \frac{2\sqrt{1-\rho^2}}{\pi} \exp\Big(-\big((1-\rho)u^2 + (\rho+1)v^2\big)\Big) du dv$$

再取 $u' = \sqrt{1-\rho}\, u, \quad v' = \sqrt{1+\rho}\, v$

故 $I = \iint \frac{2}{\pi} \exp\left(-(u'^2 + v'^2)\right) du' dv'$

$u' = \rho\cos\theta \ , \quad v' = \rho\sin\theta$

因此 $I = \int_{-\theta}^{\theta} \int_0^{\infty} \frac{2}{\pi} \exp(-r^2)\, r dr d\theta = \frac{1}{\pi}\int_{-\theta}^{\theta} d\theta = \frac{2\theta}{\pi}$

而其中： $\theta = \arctan\sqrt{\frac{1+\rho}{1-\rho}}$

从而 $I = \frac{2}{\pi}\arctan\sqrt{\frac{1+\rho}{1-\rho}}$

利用万能公式： $\cos(2\alpha) = \frac{1-\tan^2(\alpha)}{1+\tan^2(\alpha)} \Rightarrow \frac{1-\left(\sqrt{\frac{1+\rho}{1-\rho}}\right)^2}{1+\left(\sqrt{\frac{1+\rho}{1-\rho}}\right)^2} = -\rho$

故有： $I = \frac{1}{\pi}\arccos(-\rho)$

又利用 $\frac{\pi}{2} = \arcsin(x) + \arccos(x) \Rightarrow I = \frac{1}{\pi}\left(\frac{\pi}{2} - \arcsin(-\rho)\right)$

$$= \frac{1}{2} + \frac{1}{\pi}\arcsin(\rho)$$

最终有： $R_Y(t,s) = \frac{2}{\pi}\arcsin(\rho).$

Price Theorem：

$(x_1, x_2) \sim N(0, 0, \sigma_1, \sigma_2, \rho)$ , $\quad g(x,y)$ 为 Nonlinear.

有 $$\frac{\partial E(g(x_1,x_2))}{\partial\rho} = \sigma_1\sigma_2 E\left(\frac{\partial^2 g}{\partial x_1\partial x_2}(x_1,x_2)\right)$$

其中：
1. $g(x_1,x_2) = \mathrm{sgn}(x_1)\,\mathrm{sgn}(x_2)$
2. $\frac{\partial^2 g}{\partial x_1\partial x_2} = 4\delta(x_1)\delta(x_2)$
3. $\sigma_1\sigma_2 \int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty} \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}}\, 4\delta(x_1)\delta(x_2) \exp\left(-\frac{1}{2(1-\rho^2)}\left(\left(\frac{x_1}{\sigma_1}\right)^2 + \left(\frac{x_2}{\sigma_2}\right)^2 - 2\rho\frac{x_1}{\sigma_1}\frac{x_2}{\sigma_2}\right)\right) dx_1 dx_2$

$$= \frac{2}{\pi}\frac{1}{\sqrt{1-\rho^2}}$$